



1a) (5, -15) (0, -5)

$$dx = 0 - 5 = -5$$

$$dy = -5 + 15 = 10$$

Zone 2

Start (-15, -5)

End (-5, 0)

$$dx = 10$$

$$dy = 5$$

$$d_{NE} = -10$$

$$d_E = 10$$

$$d_{init} = 10 - 10$$

$$= 0$$

x	y	d	NE/E	d _{update}	Pixel ₍₂₎
-15	-5	0	E	10	(5, -15)
-14	-5	10	NE	0	(5, -14)
-13	-4	0	E	10	(4, -13)
-12	-4	10	NE	0	(4, -12)
-11	-3	0	E	10	(3, -11)
-10	-3	10	NE	0	(3, -10)
-9	-2	0	E	10	(2, -9)
-8	-2	10	NE	0	(2, -8)
-7	-1	0	E	10	(1, -7)
-6	-1	10	NE	0	(1, -6)
-5	0				(0, -5)

b) $m = \frac{10}{-5} = -2$ $\frac{1}{m} = -0.5$

$(+\frac{1}{m})x$	$(+1)y$	x (rounded)
5	-15	(5, -15)
4.5	-14	(5, -14)
4	-13	(4, -13)
3.5	-12	(4, -12)
3	-11	(3, -11)

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5	-15	(5, -15)
4.5	-14	(5, -14)
4	-13	(4, -13)
3.5	-12	(4, -12)
3	-11	(3, -11)
2.5	-10	(3, -10)
2	-9	(2, -9)
1.5	-8	(2, -8)
1	-7	(1, -7)
0.5	-6	(1, -6)
0	-5	(0, -5)

2a) 82,94,400 b) 16.67ms

c) GPU in 17ms $\rightarrow$ 8,33,333

but needed 82,94,400

so CANNOT

3a) $x_{\min} = -50$ $x_{\max} = 60$ $(-50, -70)$ $(40, 100)$
 $y_{\min} = -10$ $y_{\max} = 100$

$01 = 0100$

$02 = 0000$

$m = \frac{170}{90} = \frac{17}{9}$

$y = y_{\min} = -10$

4.5	-14	(5, -14)
4	-13	(4, -13)
3.5	-12	(4, -12)
3	-11	(3, -11)
2.5	-10	(3, -10)
2	-9	(2, -9)
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2a) 82,94,400 b) 16.67ms

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3a) $x_{\min} = -50$ $x_{\max} = 60$ $(-50, -70)$ $(40, 100)$
 $y_{\min} = -10$ $y_{\max} = 100$

$O1 = 0100$
 $O2 = 0000$

$$m = \frac{170}{90} = \frac{17}{9}$$

$$y = y_{\min} = -10$$

$$x = x_0 + \frac{1}{m} (y_{\min} - y_0)$$

$$= -50 + \frac{9}{17} (-10 + 70) = -18.24$$

New $O1 \rightarrow \underline{0000}$

b) $\rightarrow$ Used for rejection detecting

$\rightarrow$ Same place bit 1 will generate 1

$\rightarrow$ Indicating both end points in same outside region (example)